H-B-001409 (GSA-FSS)
June 5, 1968

INTERIM FEDERAL SPECIFICATION

BRUSH, CLEANING, SPOKE

This Interim Federal Specification was developed by the Standardization Division, Federal Supply Service, General Services Administration, Washington, D. C. 20405, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION.

1.1 Scope. This specification covers the requirements of spoke brushes, with natural fiber bristles and a wood handle with ferrule, attached to a twisted wire stem.

1.2 Classification. The brushes covered by this specification shall be of one size and of the following types(see 6.1).

Type 1 - 100 per cent good quality white tampico fiber.
Type 2 - 75 per cent good quality white tampico fiber, and 25 per cent oiled palmetto fiber.

2. APPLICABLE DOCUMENTS.

2.1 The following specifications and standards, of the issues in effect on date of invitation for bid or request for proposal, form a part of this specification, to the extent specified herein.

Federal Specifications:

QQ-C-320 — Chromium plating (electrodeposited).
QQ-N-290 — Nickle plating (electrodeposited).
QQ-P-416 — Plating, cadmium (electrodeposited).
QQ-Z-325 — Zinc coating, electrodeposited, requirements for.
TT-W-572 — Wood-preservative; water-repellent.
PPP-B-566 — Boxes, folding, paperboard.
PPP-B-636 — Boxes, fiberboard.
PPP-B-665 — Boxes, paperboard, metal stayed(including stay material).
PPP-B-676 — Boxes, setup.

Federal Standards:

Fed. Std. No. 123 - Marking for domestic shipment (civilian agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 7920

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Military Specifications:

MIL-W-6110 — Wood: Determination of moisture content of.

Military Standards:

MIL-STD-105 — Sampling procedures and tables for inspection by attributes.
MIL-STD-129 — Marking for shipment and storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS.

3.1 Preproduction sample. Unless otherwise specified (see 6.1), before production is commenced, a sample of the brush to be supplied shall be submitted or made available to the contracting officer or his authorized representative for examination and approval in accordance with 4.2. Approval of the preproduction sample authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Materials.

3.2.1 Wood. Wood for the handle shall be thoroughly-seasoned close-grained and straight-grained hardwood, such as beach, birch, maple or ash. The wood shall be thoroughly seasoned so that when tested in accordance with 4.3.2.1 it shall have a moisture content of not more than 12 per cent.

3.2.2 Fibers. The fibers for each brush shall be in accordance with the type specified in 1.2.

3.2.3 Wire. The wire shall be not less than 0.105 inches in diameter before any corrosive resistant treatment, and shall be made from a metal which is inherently corrosive resistant or steel which has been coated to resist corrosion. Coating shall be either cadmium in accordance with type III, class C of QQ-P-416 or zinc in accordance with type III, class 3 of QQ-Z-325, and shall show no evidence of corrosion when tested as described in 4.4.1.

3.2.4 Ferrule. The ferrule shall be made from a metal which is inherently corrosive resistant or steel which has been coated to resist corrosion. Coating shall be either cadmium in accordance with type III, class 3, of QQ-Z-325; chromium plated to a thickness of at least 0.00001 inches in accordance with types I and II, classes 1 and 2 of QQ-C-320 or nickel plated in accordance with type FS, classes 1 and 2, of QQ-N-290; and shall show no evidence of corrosion when tested as described in 4.4.1.

3.3 Construction.

3.3.1 Illustration. The illustration shown in figure 1, is for the purpose of illustrating general design and is not intended to preclude the purchase of items which are otherwise in accordance with the requirements of this specification.

3.3.2 Wood handle. The wood handle shall be a one piece spoke type design, made from wood specified in 3.2.1, one inch in diameter ± 1/8 inch, and four inches long ± 1/4 inch. A centrally located hole shall be drilled through the length of the handle. A slot or groove shall be machined in the end of the handle, parallel to its axis, to receive and secure a 3/4 inch length of the brush wire. The depth of the groove shall be such as to permit no protrusion of the secured brush wire beyond the surface of the handle. The brush end of the handle may be undercut to accommodate the ferrule. Evidence of decay, knots checks, splits or slivers is not permitted.

3.3.2.1 Finish. After all machining, the handles shall be sanded or tumbled smooth and at the option of the manufacturer, the handles shall be coated either with a clear organic coating or tumbled waxed in accordance with the manufacturers best commercial practice. The handle shall be water resistant in accordance with T-T-W-572.

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3.3.3 Brush wire. The brush wire shall be twisted with the fiber so as to allow a loop in a diameter of approximately 1/4 inch to be formed at the outer end of the brush. The two ends of the wire shall both be twisted down the length of the brush and pass through the ferrule into the drilled hole in the handle. The looped wires shall extend into the handle not less than 2 inches. One wire shall pass through the entire handle and extend for a distance of approximately 3/4 inch beyond. The 3/4 inch portion shall be bent along the side of the handle and fitted into a groove (3.3.2), so that the wire shall not extend beyond the surface of the handle.

3.3.4 Brush head. The brush head fibers (3.2.2) shall be evenly distributed and securely spiral-wound within the twisted wire core at an angle of 90 degrees to the wire. The brush bristle shall measure 4 inches in diameter at the handle part of the brush ± 1/8 inch, and shall extend parallel to the brush axis, a distance of 6 inches ± 1/4 inch. The brush shall than taper to the tip, terminating in a diameter of 2 inches ± 1/8 inch. The length of the bristle portion of the brush shall measure 10 inches overall ± 1/4 inch. There shall be a minimum of 24 twists in the 10 inch length of the brush.

3.4 Marking for identification. Unless otherwise specified (see 6.1) in addition to any special marking required by the contract or order, the manufacturer shall permanently and legibly indent the Federal Stock Number and his name or trademark of such known character easily identifiable with said manufacturer, on each brush handle.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection. The preproduction sample shall be examined for defects in 4.3.3.1.

4.3 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.3.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used are manufactured, tested, and inspected to insure conformance with the requirements of this specification.

4.3.2 In process inspection.

4.3.2.1 Moisture content of handles (wood). The handle shall be tested for moisture content prior to being machined and finished in accordance with MIL-W-6110 to determine compliance with 3.2.1. The sample unit shall be one handle. The lot shall be the number of handles being manufactured under essentially the same conditions and to be offered for inspection at one time. Three determinations shall be taken on each sample unit and the results shall be the average of three determinations. The inspection level shall be S-4 with an acceptable quality level (AQL) of 4.0 expressed in terms of defects per hundred units.

4.3.3 End item inspection.

4.3.3.1 Examination of the end item shall be in accordance with the classification of defects, inspection levels and acceptable quality levels set forth below. The lot size for the purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of brushes for examination under 4.3.3.1.1 and 4.3.3.1.2.

4.3.3.1.1 Examination of the end item for defects in appearance, construction, dimensions, and workmanship. The inspection level shall be I with an AQL of 2.5. The sample unit for this examination shall be one brush.

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TABLE I. Visual and dimensional examination

Examine	Defect
Material	
Fiber	Not of specified types. Not of specified length.
Wire	Not made of metal inherently corrosive resistant, or of steel which has been coated to resist corrosion. Less than specified minimum diameter.
Wood	Not close, straight grained hard wood.
Construction	
Fiber	Not evenly distirbuted in brush head; not uniformly and firmly held in place; can be removed by hand. Not of proper dimensions, length of fibers not in accordance with specification.
Wire	Does not contain minimum number of twists. Loop required at end of brush, not approximately 1/4 inch in diameter. Both wires do not enter handle. No wire extends through handle and beyond. Secured wire protrudes beyond surface of handle.
Wood handle	Not of diameter and length specified. Hole through center, not centrally located. Groove required for wire not of at least minimum size to accept single strand of brush wire. Evidence of decay, knots, checks, splits or slivers.
Dimensions	Any dimension not in accordance with specification requirements for size brush specified.

4.3.3.1.2 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing, and marking as required by section 5 of this specification are complied with Defects shall be scored as specified in table II. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.3.4 Testing of end items. The end item shall be tested for the applicable characteristics as specified in 4.4. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

4.4 Tests.

4.4.1 Corrosion test. Completed parts when fabricated of ferrous metal shall be placed in a boiling 10 per cent (by weight) aqueous solution of sodium chloride for a period not less than 15 minutes. The part upon being removed from the solution shall be immediately immersed for at least 1 hour in 10 per cent (by weight) aqueous solution of sodium chloride at room temperature of 68 to 85 degrees F. The part shall then be removed from this solution and, without having the adhering liquid wiped off, allowed to dry for at least 24 hours at room temperature (68 to 85 degrees F.) and then wiped free of any residue and examined for presence of corrosion. Examine parts for compliance with 3.2.3 and 3.2.4.

5. PREPARATION FOR DELIVERY.

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.1).

5.1.1 Level A and B. Twelve brushes of one type shall be packaged in a box conforming to PPP-B-566, or PPP-B-665, or PPP-B-676. The brushes shall be arranged in an alternating bristle to handle order.

5.1.2 Level C. Brushes shall be packaged in accordance with the suppliers commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

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5.2.1 Level A. Twelve unit packages shall be packed in a close fitting box conforming to PPP-B-636, class weather resistant. The box shall be closed in accordance with the appendix of the box specification.

5.2.2 Level B. Twelve unit packages shall be packed as specified in 5.2.1 except the box shall be class domestic.

5.2.3 Level C. Brushes shall be packed to insure carrier acceptance and safe delivery at destination at the lowest rates, in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the unit package and shipping container shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to marking required by the contract or order, the unit package and shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES.

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type required (see 1.2).
(c) When preproduction sample is required (see 3.1).
(d) Whether marking of brushes is required (see 3.4).
(e) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

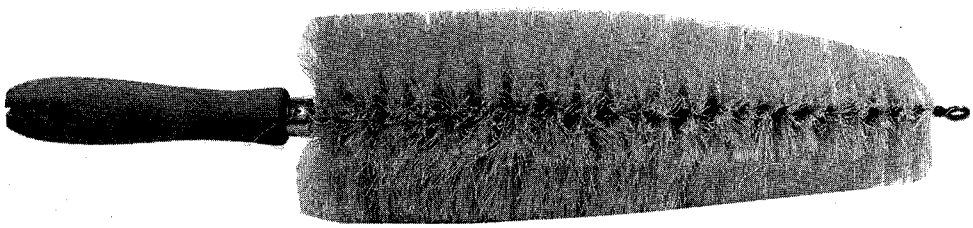


FIGURE 1.

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